Week 14 Self-Assessment

t-Tests, p-Values, and Hypothesis Testing

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1 Instructions

Test your understanding of t-tests, p-values, and the logic of hypothesis testing. These concepts form the foundation of inferential statistics in psychology research.

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2 Part 1: Conceptual Questions

2.1 Question 1: The Logic of Hypothesis Testing

We test the null hypothesis (H_0) because:

- (A) The null hypothesis is what we want to prove
- (B) We can calculate the probability of our data if the null were true
- (C) The null hypothesis is always correct
- (D) It is easier than testing the alternative hypothesis

The null hypothesis typically states:

- (A) There is a significant effect
- (B) There is no effect or no difference
- (C) Our prediction is correct
- (D) The sample is representative

2.2 Question 2: What is a p-Value?

A p-value tells us:

- (A) The probability that the null hypothesis is true
- (B) The probability that our hypothesis is correct
- (C) The probability of obtaining results this extreme (or more) if the null hypothesis were true
- (D) The size of the effect

A p-value of .03 means:

- (A) There is a 3% chance the null hypothesis is true
- (B) There is a 3% probability of getting these results if there were no real effect
- (C) 3% of participants showed the effect

- (D) The effect is 3%

2.3 Question 3: Statistical Significance

The conventional alpha level in psychology is:

- (A) .01
- (B) .05
- (C) .10
- (D) .50

We reject the null hypothesis when:

- (A) $p > .05$
- (B) $p < .05$
- (C) $p = .05$ exactly
- (D) The effect size is large

A result with $p = .07$ is:

- (A) Statistically significant
- (B) Not statistically significant at alpha = .05
- (C) Definitely not a real effect
- (D) A large effect

2.4 Question 4: Types of t-Tests

An independent samples t-test is used when:

- (A) The same participants are measured twice
- (B) Different participants are in each condition
- (C) We are measuring a correlation
- (D) Data are not normally distributed

A paired samples t-test is used when:

- (A) The same participants are measured twice (or matched pairs)
- (B) Different participants are in each condition
- (C) We have three or more groups
- (D) Variables are categorical

2.5 Question 5: One-Tailed vs Two-Tailed Tests

A one-tailed hypothesis:

- (A) Tests for any difference
- (B) Predicts a specific direction of effect
- (C) Is always more powerful
- (D) Should never be used

A two-tailed hypothesis:

- (A) Is weaker than one-tailed
- (B) Tests for a difference in either direction
- (C) Requires a larger sample
- (D) Is inappropriate for t-tests

When should you use a one-tailed test?

- (A) When you want a significant result
- (B) When theory strongly predicts a specific direction and the opposite would be meaningless
- (C) Always
- (D) Never

2.6 Question 6: t-Test Assumptions

Which assumption applies to both independent and paired t-tests?

- (A) Equal group sizes
- (B) Data should be approximately normally distributed
- (C) Perfect random sampling
- (D) Correlation between conditions

The independent t-test also assumes:

- (A) Same participants in both groups
- (B) Homogeneity of variance (similar SDs in both groups)
- (C) Exactly equal means
- (D) Large effect sizes

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3 Part 2: Interpretation Questions

3.1 Question 7: Reading t-Test Output

A t-test output shows: $t(38) = 2.45$, $p = .019$

The degrees of freedom are:

- (A) 2.45
- (B) 38
- (C) .019
- (D) Cannot tell

Is this result significant at alpha = .05? TRUE / FALSE

The number in parentheses (38) indicates:

- (A) Sample size
- (B) Degrees of freedom
- (C) Effect size

- (D) Group size

3.2 Question 8: Interpreting Direction

Results show: $t(28) = -3.21$, $p = .003$

The negative t-value indicates:

- (A) The result is not significant
- (B) An error in calculation
- (C) The second mean was higher than the first
- (D) The effect size is negative

This result is:

- (A) Not significant
- (B) Statistically significant
- (C) Inconclusive
- (D) Invalid

3.3 Question 9: Effect Size with t-Tests

Cohen's d is reported alongside t-tests to indicate:

- (A) Whether the result is significant
- (B) The magnitude of the difference in standard deviation units
- (C) The probability of the null hypothesis
- (D) The reliability of the measure

If $t(40) = 2.10$, $p = .042$, $d = 0.65$, the effect size is:

- (A) Small
- (B) Medium
- (C) Large
- (D) Not significant

3.4 Question 10: Degrees of Freedom

For an independent t-test with 25 participants per group (50 total), df =

- (A) 50
- (B) 48
- (C) 25
- (D) 49

For a paired t-test with 30 participants, df =

- (A) 30
- (B) 29
- (C) 28
- (D) 60

4 Part 3: Application Questions

4.1 Question 11: Choosing the Right Test

Scenario A: A researcher compares test anxiety between students who received study skills training versus a control group (different students in each condition).

Appropriate test:

- (A) Independent samples t-test
- (B) Paired samples t-test
- (C) Pearson's correlation
- (D) Chi-square

Scenario B: A researcher measures depression scores before and after a therapy intervention in the same patients.

Appropriate test:

- (A) Independent samples t-test
- (B) Paired samples t-test
- (C) Pearson's correlation
- (D) One-sample t-test

Scenario C: A researcher compares reaction times between morning people and evening people (different participants).

Appropriate test:

- (A) Independent samples t-test
- (B) Paired samples t-test
- (C) One-sample t-test
- (D) Correlation

4.2 Question 12: Formulating Hypotheses

A researcher predicts that caffeine will improve alertness.

One-tailed hypothesis:

- (A) Caffeine will affect alertness
- (B) Participants who consume caffeine will have higher alertness than those who do not
- (C) There is no effect of caffeine
- (D) Alertness varies

Two-tailed hypothesis:

- (A) There will be a difference in alertness between caffeine and no-caffeine conditions
- (B) Caffeine increases alertness
- (C) Caffeine decreases alertness
- (D) Caffeine has no effect

Null hypothesis:

- (A) Caffeine improves alertness
- (B) There is no difference in alertness between conditions
- (C) Caffeine affects alertness somehow
- (D) The sample is biased

4.3 Question 13: Interpreting Non-Significant Results

A study finds: $t(48) = 1.45$, $p = .154$, $d = 0.41$

Is this statistically significant at alpha = .05? TRUE / FALSE

Does $p = .154$ prove the null hypothesis is true? TRUE / FALSE

The effect size ($d = 0.41$) suggests:

- (A) No effect at all
- (B) A small-to-medium effect that may warrant further investigation
- (C) Proof that the treatment works
- (D) The study was flawed

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5 Part 4: MCQ Exam Practice

5.1 Investigation 1

A researcher tests whether a memory training app improves recall. 40 participants are randomly assigned to use the app or a control app for 2 weeks. Memory is tested with a word recall task.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Matched pairs

2. **IV:**

- (A) App type (memory training vs control)
- (B) Recall score
- (C) Training duration
- (D) No IV

3. **DV:**

- (A) App condition
- (B) Word recall score
- (C) Time spent training
- (D) Participant age

4. **Appropriate test:**

- (A) Independent samples t-test
- (B) Paired samples t-test
- (C) Pearson's r
- (D) Chi-square

5. **Results: $t(38) = 2.78$, $p = .008$, $d = 0.88$. Is this significant at alpha = .05?** TRUE / FALSE

6. **Effect size interpretation:**

- (A) Small
- (B) Medium
- (C) Large
- (D) Cannot determine

5.2 Investigation 2

A researcher examines whether mood changes after watching a sad film. 25 participants rate their mood before and after watching the film.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Correlational
- (D) Case study

2. **IV:**

- (A) Time (before vs after film)
- (B) Mood rating
- (C) Film content
- (D) No IV

3. **DV:**

- (A) Time point
- (B) Mood rating
- (C) Film preference
- (D) Participant enjoyment

4. **Appropriate test:**

- (A) Independent t-test
- (B) Paired t-test
- (C) Correlation
- (D) ANOVA

5. **Results: $t(24) = -4.12$, $p < .001$, $d = 0.82$. The negative t-value indicates:**

- (A) The result is not significant
- (B) Mood decreased after the film
- (C) An error occurred
- (D) Mood increased

6. **Conclusion:**

- (A) Mood was significantly lower after watching the sad film, large effect
- (B) No change in mood
- (C) The study was underpowered
- (D) Mood improved

5.3 Investigation 3

A researcher compares stress levels between employees who work from home versus in the office. 30 work-from-home employees and 30 office employees complete a stress questionnaire.

1. **Design type:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Repeated measures
- (D) Correlational

2. **IV:**

- (A) Work location (home vs office)
- (B) Stress level
- (C) Employee satisfaction
- (D) No IV

3. **DV:**

- (A) Work location
- (B) Stress score
- (C) Job type
- (D) Hours worked

4. **Degrees of freedom would be:**

- (A) 60
- (B) 58
- (C) 30
- (D) 29

5. **Results: $t(58) = 1.89$, $p = .064$, $d = 0.49$. Is this significant at alpha = .05?** TRUE / FALSE

6. **The effect size suggests:**

- (A) No effect
- (B) A medium effect that may be worth investigating with more power
- (C) A large, meaningful effect
- (D) The hypothesis was wrong

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6 Summary Checklist

Before moving on, make sure you can:

- ☐ Explain the logic of null hypothesis significance testing
- ☐ Correctly interpret p-values (NOT as the probability the null is true)
- ☐ Distinguish between independent and paired t-tests
- ☐ Choose the appropriate t-test for different designs
- ☐ Interpret t-test output including degrees of freedom
- ☐ Write one-tailed and two-tailed hypotheses
- ☐ Understand what non-significant results do and do not tell us
- ☐ Report t-test results in APA format

> ! Common Misconception
>
> A p-value is NOT the probability that the null hypothesis is true. It is the probability of obtaining your results (or more extreme) IF the null hypothesis were true. This distinction is crucial for correctly interpreting statistical tests.